INTRODUCTION

We already know that proteins are made up of amino acids and have many functions in all living system. The 20 amino acids are obtained from the diet or from turnover of proteins. The major part of amino acid content of the cell exists in the form of proteins which is constantly being synthesized and degraded. Amino acids cannot be stored in an analogous form of glycogen (glucose) or triglycerides (fatty acids) leaving a small pool of amino acids in the cells as per the needs for protein synthesis. Excess amino acids are broken down into the carbon skeleton and the nitrogen amino group. The carbon skeleton is converted into common metabolites that can be used for the synthesis of other biomolecules or for further oxidation to generate ATP, while their amino group is converted to urea.

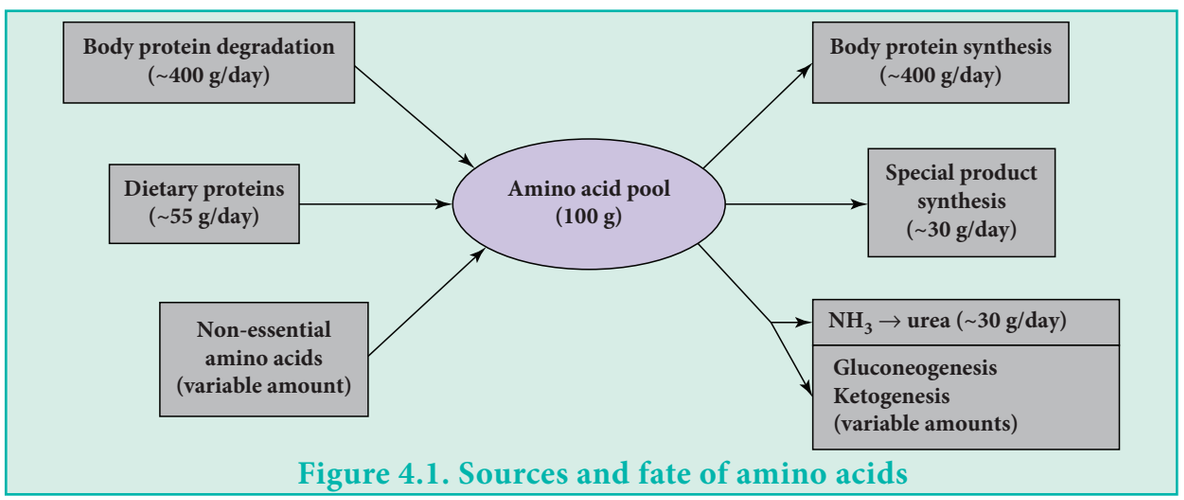

Figure 4.1. Sources and fate of amino acids

Amino acids are not only used to synthesize proteins, but also they act as precursors for specialized products of amino acid derivatives that play biologically significant roles in metabolism and homeostasis (Figure 4.1).

4.1 GENERAL REACTIONS OF AMINO ACID

Non-essential amino acids are synthesized from α-keto acids by transfer of amino groups with the help of transaminases. However, essential amino acids cannot be synthesized from α-keto acids. Transfer of amino groups also occurs during degradation of amino acids. Amino groups can be removed from amino acids by transamination as well as by deamination. Removal of carboxyl group (Decarboxylation) from amino acids results in the production of biologically active amines. The carbon skeletons of amino acids are degraded on entry into TCA cycle.

4.1.1 Catabolism of amino acid

Even though, each amino acid has a specific pathway for its catabolism, the first step in the catabolism of amino acids is the removal of its amino group. This can happen by two different ways namely, transamination and deamination.

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